

geous. Customising the Linux kernel by removing unwanted drivers creates a compact footprint as well as tighter and verifiable codes.

4. Implement the power management function specifically for your hardware by measuring the power consumed and understand the battery that is being used. Many a time, using a generic power management function results in sub-optimal performance. Knowledge of processor core and the memory used is important for implementing the power management function. A wearable device's success mostly depends on its battery life.

5. Ensure that the software is upgradable and also has sufficient headroom for code expansion in the hardware. It is advisable to have only about 70 per cent memory to be occupied when releasing the product. When the memory usage reaches 90 per cent, you can revise product hardware with a higher capacity.

6. Since most wearable devices store data before these are shipped out, ensuring a reliable mechanism that occupies optimal space to store data is important. For example, if the position of 1-switch data uses a byte, packing 8-switch data in one byte can save huge memory area.

7. C based application is suitable for real-time and critical time-sensitive applications. With proliferation of Java, designers tend to use it but for power-sensitive and performance-critical applications, languages like C are better suited.

Development process

While lack of a process may be fine when the product volumes are small, as volumes increase it is risky not to have a process in place, especially in the case of wearable devices. Since most wearable devices interact with humans, and humans use these in their daily regimen, having design decisions documented by following a process will help avoid any future litigation. Let us see an example.

You have designed a wearable

device where you have calculated the battery life of 2.5 years and you certify that the device will work for two years before requiring a replacement. If a user continues to use this, let us say for three years, and the device fails and the user sues the company, you should be able to show your documentation to prove that your device is meant to work for two years and at best 2.5 years, and anything beyond is not guaranteed.

Most medical devices have a well-documented process defined by ISO 13485 standard, and design process documents are stored till the product is withdrawn from the market. These files are known as design history files. The medical device development process is exhaustive and takes time and effort to create and maintain. While wearable devices need not have such a detailed process, there are a few documents that when created will help designers and the company to manage the product successfully.

Given below are the essential documents that designers need to create and maintain for a successful wearable device:

Product requirement document. This document captures every product feature that will be implemented. Every feature is uniquely numbered so that each requirement can be tested and validated.

Material selection document. This document captures the details of all materials considered for product packaging and their characteristics, and the material chosen and technical reasons for the same.

System architecture document. This document covers product architecture at the highest level (as block diagrams), the specific device chosen and the reason for that choice. This should also cover software architecture and software components chosen and the reason for choosing these.

System safety analysis document. This document analyses the

safety features and the impact on the user and its safety. Let us say, due to some hardware problem the wearable device starts reading double heartbeat as normal. In this case, the software should have a feature that compares the reading with standard set limits and, if data is out of bound, raises an alarm. This comes in handy when the wearable measures body parameters and users use it for their daily health monitoring. So if wrong data is read by the system, it should not force users to change their daily routine due to this error.

Critical design decision document. This document captures all critical design decisions covering selection of parts to the way PCB has been laid out, test strategy and test hooks, among others. This document comes in handy when the next-generation product has to be designed or when product revision has to be done.

Test strategy document. This is a key document where the test strategy adopted for testing the PCB, product and calibration procedure and details of the tester are recorded. It helps in managing the yield as well as improving productivity. Above all, in case of failure, this document ensures isolation of the problem found in the manufacturing line.

Product master record. This document captures the product's serial numbers and corresponding test results with the measurement so that when failure due to aging happens, the reasons can be tracked back to the batch and the process involved. It is typically created out of manufacturing line and, during contract negotiation, data capture has to be put in contract.

The primary idea behind this article was to demystify wearable devices and the essential characteristics these should have. While this article does not serve as a comprehensive guide, it does help to understand the intricacies of a wearable device design and how it can be designed for success. **EFY**

Concluded